

program, which is not predictable, and hence it has a dynamic lifetime.

Visibility

Visibility is the 'accessibility' of the variable declared. It is the result of hiding a variable in outer scopes. Here is a dummy example:

```
int i;
// the "i" variable is accessible/visible here
void foo() {
    int i;
    // the outer "i" variable is not
    // accessible/visible here
    {
        int i;
        // two "i" variables at outer scopes
        // are not accessible/visible here
    }
    // the "i" in this block is accessible/visible
    // here and it still hides the outer "i"
```

```
}
// the outermost "i" variable is accessible/visible here
```

Summary of differences

As you can see, scope, lifetime and visibility are related to each other, but are distinct. Scope is about the 'availability' of the declared variable: within the same scope, it is not possible to declare/define two variables of the same type with the same name. Lifetime is about the duration in which the variable is 'alive': it determines how long the named or unnamed variable has memory allocated to it. Visibility is about the 'accessibility' of the declared variables: it arises because of the possibility of variables in outer scope having the same name as the ones in inner scopes, resulting in 'hiding'. 

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Go Online to learn Linux Device Drivers



Core skills like kernel programming or Linux device drivers should be learned at a slow and steady pace for fully comprehending it, says Mr. Raghu Bharadwaj, a lead trainer on core Linux programming. Unlike normal application software learning where it is mostly restricted to functionality based learning, system programming involves deep understanding and analysis of issues like the Linux kernel and its subsystems. Your options to learn these skills are classroom based training, learning through experience, self learning through trial and error and through an effective online course.

While all other modes of training has its own merits and demerits, online learning is definitely a step ahead, especially for courses like this. Online courses gives you the comfort of accessing it 24x7 and lets you learn things during hours of focus and interest. Another important aspect is that it permits you to go through each session any number of times and fully understand each and every topic effectively.

Mr. Raghu Bharadwaj is a lead trainer at Veda Solutions, you can listen to his **training videos** on Linux kernel and device drivers at www.techveda.org

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